

Experiment Catalogue

INFORMS 2026 DMS Data Challenge

All 210 scored out-of-fold runs, classified by the *contrast* each tested. County-holdout cross-validation, 239 counties, five folds. RMSE is the mean of the four per-horizon values (a reporting summary; the organisers rank within each horizon).

Summary by intervention family

code	family	runs	best	median	shipped
F1	Signal	45	0.00930	0.00963	1
F2	Mechanism	67	0.00893	0.01042	3
F3	Target	27	0.00863	0.00960	4
F4	Exposure	27	0.00893	0.00955	0
F5	Estimation	23	0.00862	0.00905	3
F6	Adaptation	7	0.00892	0.00938	1
F7	Combination	4	0.00920	0.01149	0
F8	Uncertainty	1	0.01032	0.01032	0
P	Protocol	4	0.00914	0.00923	0
X	Compound	5	0.00964	0.01086	0
total		210	0.00862	0.00973	12

F1 — Signal (45 runs)

Contrast tested: Change what information the predictor may see.

Prediction from the diagnosis: The residual multiplier is not recoverable from legal features, so added information should return little.

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
F2	s45 pipeline + interactions (member upgrade)	boosted / bagged trees	sqrt target	0.00930	0.01145	0.01011	0.00816	0.00746	0.00254	17
FS2e	full FE + shadow (Boruta-lite)	boosted / bagged trees	plain squared error	0.00934	0.01144	0.01017	0.00823	0.00753	0.00250	13
FS2c	full FE + permutation importance	boosted / bagged trees	plain squared error	0.00935	0.01141	0.01011	0.00824	0.00763	0.00253	14
F1d	interactions only	boosted / bagged trees	sqrt target	0.00935	0.01151	0.01020	0.00824	0.00746	0.00252	43
F1	independent-agent features (pre-event frag, clocks, inte	boosted / bagged trees	sqrt target	0.00938	0.01155	0.01019	0.00822	0.00756	0.00253	46
F4	f2 + decay-modifier interactions	boosted / bagged trees	sqrt target	0.00941	0.01173	0.01024	0.00817	0.00750	0.00252	9
S45	sub-county gust heterogeneity	Seq2Seq (GRU)	plain squared error	0.00942	0.01148	0.01023	0.00832	0.00766	0.00252	8
#1										
FS2b	full FE + gain top-40	boosted / bagged trees	plain squared error	0.00944	0.01166	0.01031	0.00822	0.00760	0.00255	9
S46b	sub-county scoring-aligned bi-enc	BiEncSeq2Seq	plain squared error	0.00951	0.01129	0.01011	0.00866	0.00799	0.00307	133

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
E4	sqrt-XGB + EIA meters + elev_range	boosted / bagged trees	sqrt target	0.00954	0.01166	0.01035	0.00839	0.00774	0.00255	16
F1e	veer only	boosted / bagged trees	sqrt target	0.00954	0.01163	0.01038	0.00837	0.00779	0.00253	47
S60d	sqrt-XGB + statics + sub-county	boosted / bagged trees	sqrt target	0.00956	0.01161	0.01042	0.00843	0.00780	0.00253	26
S47	BiGRU + external county data	Seq2Seq (GRU)	plain squared error	0.00956	0.01116	0.01014	0.00880	0.00815	0.00309	130
S46	sub-county bi-encoder	Seq2Seq (GRU)	plain squared error	0.00958	0.01113	0.01013	0.00881	0.00824	0.00313	128
E4c	sqrt-XGB + HIFLD transmission	boosted / bagged trees	sqrt target	0.00958	0.01172	0.01037	0.00843	0.00779	0.00257	12
S60b	sqrt-XGB + statics	boosted / bagged trees	sqrt target	0.00960	0.01175	0.01045	0.00840	0.00779	0.00251	25
U1c	control rerun (statics only)	boosted / bagged trees	sqrt target	0.00960	0.01175	0.01045	0.00840	0.00779	0.00251	10
U1	sqrt-XGB + statics + true URMA field	boosted / bagged trees	sqrt target	0.00961	0.01171	0.01043	0.00850	0.00780	0.00258	12
F1b	clocks only	boosted / bagged trees	sqrt target	0.00962	0.01177	0.01048	0.00844	0.00779	0.00256	42
S61	+ spatial storm-exposure summaries	boosted / bagged trees	sqrt target	0.00962	0.01171	0.01042	0.00850	0.00787	0.00256	32
E4b	sqrt-XGB + 4 sanctioned cols	boosted / bagged trees	sqrt target	0.00962	0.01180	0.01046	0.00844	0.00780	0.00254	14
S45	sub-county gust heterogeneity	Seq2Seq (GRU)	plain squared error	0.00963	0.01167	0.01041	0.00851	0.00790	0.00255	8
#2										
S47b	BiGRU + ext + scoring-aligned	BiEncSeq2Seq	plain squared error	0.00963	0.01139	0.01025	0.00876	0.00813	0.00308	106
F3	f2 + winter-hazard load features	boosted / bagged trees	sqrt target	0.00968	0.01185	0.01050	0.00851	0.00785	0.00257	172
U1b	sqrt-XGB + statics + sub + URMA	boosted / bagged trees	sqrt target	0.00969	0.01181	0.01052	0.00851	0.00792	0.00254	13
F1c	pre-event fragility only	boosted / bagged trees	sqrt target	0.00974	0.01187	0.01059	0.00854	0.00796	0.00257	44
E4d	ExtraTrees + EIA meters + elev	boosted / bagged trees	sqrt target	0.00974	0.01170	0.01044	0.00868	0.00816	0.00247	37
S61b	ExtraTrees + all tabular info	boosted / bagged trees	sqrt target	0.00976	0.01167	0.01041	0.00872	0.00824	0.00245	96
I08	transductive pretraining	Seq2Seq (GRU)	plain squared error	0.00984	0.01219	0.01065	0.00857	0.00797	0.00322	201
T1	transductive multi-origin (test coun- ties' own 72h)	Seq2Seq (GRU)	plain squared error	0.00992	0.01147	0.01067	0.00902	0.00852	0.00283	477
FS2a	full FE + corr filter	boosted / bagged trees	plain squared error	0.00994	0.01174	0.01058	0.00903	0.00840	0.00262	7
FS1d	base + lasso	boosted / bagged trees	plain squared error	0.00996	0.01167	0.01043	0.00916	0.00860	0.00266	7
FS0	base features, no selection (control)	boosted / bagged trees	plain squared error	0.01005	0.01180	0.01070	0.00911	0.00857	0.00269	6
S60	sqrt-XGB base (no statics, no sub)	boosted / bagged trees	sqrt target	0.01005	0.01180	0.01070	0.00911	0.00857	0.00269	19
FS1e	base + shadow (Boruta-lite)	boosted / bagged trees	plain squared error	0.01009	0.01187	0.01071	0.00909	0.00868	0.00264	9
FS1b	base + gain top-40	boosted / bagged trees	plain squared error	0.01010	0.01193	0.01074	0.00907	0.00865	0.00266	6
FS2d	full FE + lasso	boosted / bagged trees	plain squared error	0.01012	0.01183	0.01073	0.00930	0.00863	0.00269	10
FS1a	base + corr filter	boosted / bagged trees	plain squared error	0.01013	0.01175	0.01074	0.00935	0.00867	0.00269	5
S60c	sqrt-XGB + sub-county	boosted / bagged trees	sqrt target	0.01016	0.01189	0.01073	0.00923	0.00879	0.00266	22
FS1c	base + permutation importance	boosted / bagged trees	plain squared error	0.01040	0.01232	0.01114	0.00929	0.00883	0.00271	11
S41	RAFT retrieved analogs	other / statistical	plain squared error	0.01045	0.01227	0.01109	0.00955	0.00888	0.00334	160
#1										
S41	RAFT retrieved analogs	other / statistical	plain squared error	0.01045	0.01227	0.01109	0.00955	0.00888	0.00334	166
#2										
I17b	static + physics	Seq2Seq (GRU)	plain squared error	0.01046	0.01294	0.01159	0.00903	0.00827	0.00333	17
I16	static enrichment	Seq2Seq (GRU)	plain squared error	0.01047	0.01274	0.01151	0.00920	0.00842	0.00334	19
I17	physics indices	Seq2Seq (GRU)	plain squared error	0.01081	0.01296	0.01172	0.00960	0.00896	0.00336	18

F2 — Mechanism (67 runs)

Contrast tested: Hold information and task fixed; change the map from covariates to forecast.

Prediction from the diagnosis: Error is spread across scale, shape and timing, so no single function class should capture it.

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
E5	2-layer bi-encoder + S62 loss	BiEncSeq2Seq	dual raw+sqrt	0.00893	0.01062	0.00966	0.00811	0.00735	0.00257	218
S31	bidirectional encoder	LSTM / Transformer	plain squared error	0.00910	0.01075	0.00966	0.00837	0.00761	0.00297	167
E3	N.t transition auxiliary head	LSTM / Transformer	plain squared error	0.00912	0.01059	0.00984	0.00840	0.00767	0.00257	161
I07	cross-county attention	CrossCountyNet	plain squared error	0.00931	0.01125	0.01005	0.00843	0.00752	0.00306	194
B3	geo-temporal adjacency GNN (2025 3rd-place analog)	other / statistical	plain squared error	0.00932	0.01112	0.01008	0.00842	0.00767	0.00265	654
S39	cross-county self-leak fixed	CrossCountyNet	plain squared error	0.00939	0.01133	0.01019	0.00848	0.00756	0.00314	203
S39b	cross-county + transductive pool	CrossCountyNet	plain squared error	0.00939	0.01134	0.01019	0.00848	0.00756	0.00314	196
E2	Transformer + dual raw/sqrt	LSTM / Transformer	plain squared error	0.00950	0.01126	0.01031	0.00853	0.00789	0.00272	572
TS2b	PatchTST-KF winsorised (Nie ICLR 2023 + known-future pat	patch / variate atten- tion	plain squared error	0.00951	0.01178	0.01055	0.00834	0.00736	0.00291	941
S30	LSTM seq2seq	LSTM / Transformer	plain squared error	0.00955	0.01146	0.01031	0.00857	0.00786	0.00301	128
S43	bagged bi-encoder (county subsets)	BiEncSeq2Seq	plain squared error	0.00957	0.01123	0.01011	0.00873	0.00820	0.00310	148
S29d	sqrt + XGBoost	boosted / bagged trees	plain squared error	0.00960	0.01175	0.01045	0.00840	0.00779	0.00251	8
S29c	sqrt + LightGBM	boosted / bagged trees	plain squared error	0.00961	0.01172	0.01045	0.00850	0.00778	0.00255	11
S23b	ExtraTrees	boosted / bagged trees	plain squared error	0.00970	0.01157	0.01038	0.00868	0.00817	0.00310	32
E1	component-GRU + S62 objective	other / statistical	plain squared error	0.00971	0.01200	0.01069	0.00852	0.00763	0.00287	131
SX2	TSMixer-Ext-KF (Chen 2023; align + conditional feature m	TCN / basis / linear	plain squared error	0.00977	0.01193	0.01064	0.00857	0.00792	0.00294	165
I12	TFT-lite	LSTM / Transformer	plain squared error	0.00982	0.01127	0.01052	0.00923	0.00826	0.00333	143
S29	sqrt + ExtraTrees	boosted / bagged trees	plain squared error	0.00987	0.01180	0.01054	0.00881	0.00834	0.00248	30
S29b	sqrt + RandomForest	boosted / bagged trees	plain squared error	0.00988	0.01187	0.01064	0.00883	0.00819	0.00256	101
D3	TabNet on the f2 feature set	MLP / KAN / TabNet / FT-T	plain squared error	0.00989	0.01141	0.01053	0.00902	0.00861	0.00250	120
D1	MLP on the f2 feature set	MLP / KAN / TabNet / FT-T	plain squared error	0.00998	0.01176	0.01101	0.00887	0.00830	0.00254	17
S29e	sqrt + bagged ExtraTrees	boosted / bagged trees	plain squared error	0.01000	0.01204	0.01071	0.00888	0.00836	0.00252	91
S23	Random Forest	boosted / bagged trees	plain squared error	0.01005	0.01198	0.01078	0.00903	0.00843	0.00328	133
D4	FT-Transformer on the f2 feature set	MLP / KAN / TabNet / FT-T	plain squared error	0.01007	0.01173	0.01084	0.00921	0.00851	0.00262	15525
K1	kernel-ridge functional output (sqrt)	TCN / basis / linear	plain squared error	0.01017	0.01299	0.01130	0.00852	0.00788	0.00264	64
TS3b	iTransformer-KF winsorised (Liu ICLR 2024 + known-future	patch / variate atten- tion	plain squared error	0.01023	0.01256	0.01134	0.00893	0.00811	0.00315	67
K1b	kernel-ridge functional output (raw)	TCN / basis / linear	plain squared error	0.01025	0.01273	0.01122	0.00882	0.00821	0.00353	53
S22c	LightGBM DART	boosted / bagged trees	plain squared error	0.01025	0.01259	0.01117	0.00893	0.00829	0.00317	27
M2	cumulative-curve reformulation (inte- grals)	TCN / basis / linear	plain squared error	0.01025	0.01194	0.01100	0.00930	0.00878	0.00313	10
S21	XGBoost	boosted / bagged trees	plain squared error	0.01028	0.01219	0.01101	0.00925	0.00867	0.00334	11

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
R8b	non-recursive stock-flow (fraction restore)	stock-flow state space	hurdle / two-part	0.01030	0.01213	0.01131	0.00927	0.00848	0.00286	13
I14	hierarchical pooling	other / statistical	plain squared error	0.01032	0.01209	0.01103	0.00952	0.00864	0.00324	70
S22	LightGBM	boosted / bagged trees	plain squared error	0.01042	0.01268	0.01137	0.00917	0.00845	0.00338	15
P1b	stock-flow hurdle (no monotone)	stock-flow state space	hurdle / two-part	0.01042	0.01221	0.01136	0.00955	0.00856	0.00447	18
#1										
S22b	LightGBM Tweedie	boosted / bagged trees	plain squared error	0.01043	0.01285	0.01119	0.00913	0.00854	0.00249	21
S27	feature selection top-40	boosted / bagged trees	plain squared error	0.01047	0.01278	0.01151	0.00918	0.00841	0.00340	11
S28	quantile-transformed features	boosted / bagged trees	plain squared error	0.01052	0.01297	0.01167	0.00911	0.00834	0.00336	27
S32	temporal CNN (TCN)	TCN / basis / linear	plain squared error	0.01054	0.01228	0.01141	0.00958	0.00890	0.00365	395
D5	proper B-spline KAN (quantile inputs, base residual)	MLP / KAN / TabNet / FT-T	plain squared error	0.01059	0.01329	0.01163	0.00908	0.00837	0.00301	772
S33	basis expansion (N-BEATS style)	TCN / basis / linear	plain squared error	0.01060	0.01225	0.01144	0.00970	0.00899	0.00375	33
S21b	XGBoost Tweedie	boosted / bagged trees	plain squared error	0.01066	0.01323	0.01163	0.00919	0.00859	0.00254	9
S34	physics-informed net	other / statistical	plain squared error	0.01071	0.01228	0.01146	0.00998	0.00913	0.00378	126
SX4	TimeXer-KF (Wang NeurIPS 2024; global-token cross-attent	patch / variate attention	plain squared error	0.01082	0.01334	0.01193	0.00937	0.00864	0.00312	89
I04	wave-conditioned cascade	other / statistical	plain squared error	0.01086	0.01270	0.01163	0.00990	0.00922	0.00346	39
P1 #1	stock-flow hurdle (monotone)	stock-flow state space	hurdle / two-part	0.01089	0.01270	0.01185	0.01000	0.00899	0.00453	19
D2	Chebyshev-KAN on the f2 feature set	MLP / KAN / TabNet / FT-T	plain squared error	0.01092	0.01406	0.01217	0.00912	0.00832	0.00383	19
S23c	Bayesian Ridge	boosted / bagged trees	plain squared error	0.01096	0.01263	0.01185	0.01002	0.00933	0.00409	5
SX1	XLinear-KF (Ma 2026; gated MLP with exogenous routing)	TCN / basis / linear	plain squared error	0.01112	0.01361	0.01211	0.00975	0.00900	0.00349	28
B2b	cluster-VARX direct form (adapted to 2026 regime)	other / statistical	plain squared error	0.01113	0.01292	0.01204	0.01009	0.00948	0.00388	2
S23d	ElasticNet	boosted / bagged trees	plain squared error	0.01115	0.01284	0.01205	0.01019	0.00952	0.00406	12
I02	impulse-response (system ID)	other / statistical	plain squared error	0.01120	0.01288	0.01223	0.01036	0.00934	0.00413	13
S42	TiDE (known-future covariates)	TCN / basis / linear	plain squared error	0.01121	0.01360	0.01229	0.00987	0.00909	0.00382	17
P1 #2	stock-flow hurdle (monotone)	stock-flow state space	hurdle / two-part	0.01148	0.01351	0.01267	0.01039	0.00934	0.00384	9
P1c	stock-flow, teacher-forced only	stock-flow state space	hurdle / two-part	0.01148	0.01351	0.01267	0.01039	0.00934	0.00384	9
P1b	stock-flow hurdle (no monotone)	stock-flow state space	hurdle / two-part	0.01158	0.01355	0.01272	0.01052	0.00951	0.00388	8
#2										
P1d	stock-flow, 3 DAgger rounds	stock-flow state space	hurdle / two-part	0.01191	0.01340	0.01264	0.01112	0.01048	0.00641	38
TS1b	DLinear-KF winsorised (Zeng AAAI 2023 + known-future)	TCN / basis / linear	plain squared error	0.01204	0.01513	0.01296	0.01033	0.00974	0.00337	8
I03	FPCA two-stage	other / statistical	plain squared error	0.01247	0.01617	0.01422	0.01043	0.00905	0.00407	5
R8	non-recursive stock-flow (gross flows)	stock-flow state space	hurdle / two-part	0.01986	0.02030	0.02005	0.01961	0.01950	0.00550	14
S26	county-normalized target	boosted / bagged trees	plain squared error	0.02164	0.02568	0.02408	0.02288	0.01392	0.00543	16
B2	cluster-VAR (2025 2nd-place analog)	other / statistical	plain squared error	0.02223	0.02271	0.02224	0.02160	0.02235	0.00906	3
TS2	PatchTST-KF (Nie ICLR 2023 + known-future patches)	patch / variate attention	plain squared error	0.02550	0.03683	0.03298	0.01625	0.01595	0.00361	1172
B1 #1	SARIMAX(2,0,2) per-county baseline	other / statistical	plain squared error	0.02775	0.02809	0.02797	0.02745	0.02749	0.00745	10

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
S25	binning classification	boosted / bagged trees	plain squared error	0.03353	0.03695	0.03431	0.03042	0.03242	0.00610	87
TS3	iTransformer-KF (Liu ICLR 2024 + known-future variates)	patch / variate attention	plain squared error	0.05797	0.06981	0.06457	0.04753	0.04996	0.00518	85
TS1	DLinear-KF (Zeng AAAI 2023 + known-future)	TCN / basis / linear	plain squared error	0.08799	0.13170	0.12663	0.06556	0.02807	0.00638	5
B1 #2	SARIMAX(2,0,2) per-county baseline	other / statistical	plain squared error	—	—	—	—	—	—	9

F3 — Target (27 runs)

Contrast tested: Change what prediction problem is optimised: loss, target scale, factorisation.

Prediction from the diagnosis: 42.4% zeros are cheap (0.7% of squared error); skew is expensive.

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
A11x25	snapshot + exact-metric, 25 seeds	BiEncSeq2Seq	dual raw+sqrt	0.00863	0.01019	0.00925	0.00790	0.00720	0.00247	8013
A11	snapshot + exact-metric weights	BiEncSeq2Seq	dual raw+sqrt	0.00869	0.01013	0.00929	0.00805	0.00731	0.00249	880
A5	S62 + exact-metric hour weights	BiEncSeq2Seq	dual raw+sqrt	0.00888	0.01031	0.00961	0.00824	0.00737	0.00253	169
X9w	s51 + snapshots + metric hour weights	BiEncSeq2Seq	dual raw+sqrt	0.00891	0.01047	0.00958	0.00819	0.00741	0.00251	487
S62	BiGRU + scoring + dual loss	BiEncSeq2Seq	dual raw+sqrt	0.00897	0.01028	0.00961	0.00835	0.00763	0.00257	440
S44b	scoring-aligned bi-encoder	BiEncSeq2Seq	metric hour weights	0.00904	0.01070	0.00966	0.00829	0.00749	0.00300	166
I11w	i11 storm augmentation + metric hour weights	Seq2Seq (GRU)	metric hour weights	0.00909	0.01062	0.00989	0.00829	0.00756	0.00276	2886
S51	BiGRU dual raw+sqrt loss	BiEncSeq2Seq	dual raw+sqrt	0.00916	0.01055	0.00985	0.00851	0.00775	0.00265	140
S35w	s35 mixup + metric hour weights	Seq2Seq (GRU)	metric hour weights	0.00930	0.01093	0.01009	0.00849	0.00767	0.00302	1605
S39w	s39 cross-county + metric hour weights	CrossCountyNet	metric hour weights	0.00944	0.01190	0.01031	0.00815	0.00739	0.00310	415
I05b	archetype ORACLE (ceiling)	boosted / bagged trees	plain squared error	0.00945	0.01205	0.01060	0.00812	0.00702	0.00313	76
TS2c	PatchTST-KF + metric hour weights	patch / variate attention	dual raw+sqrt	0.00952	0.01173	0.01054	0.00842	0.00738	0.00291	1404
TS2d	PatchTST-KF + exact metric Jacobian	patch / variate attention	dual raw+sqrt	0.00954	0.01179	0.01056	0.00841	0.00739	0.00290	1392
S54	severe-event auxiliary head	other / statistical	plain squared error	0.00960	0.01122	0.01018	0.00879	0.00819	0.00303	117
S24b	sqrt target	boosted / bagged trees	sqrt target	0.00967	0.01198	0.01058	0.00838	0.00774	0.00257	23
H2	hurdle-gated + occurrence BCE 0.1	other / statistical	hurdle / two-part	0.00968	0.01129	0.01034	0.00893	0.00816	0.00270	130
S50	BiGRU sqrt target	BiEncSeq2Seq	sqrt target	0.00971	0.01133	0.01040	0.00891	0.00821	0.00261	122
H1	hurdle-gated bi-encoder (S62 loss)	other / statistical	hurdle / two-part	0.00976	0.01144	0.01053	0.00893	0.00815	0.00269	127
S37	multi-task (OSI+P+N)	other / statistical	plain squared error	0.00984	0.01179	0.01061	0.00891	0.00806	0.00322	123
SX3	CondFlow-KF (conditional affine-coupling flow, sample-me)	normalizing flow	likelihood	0.00989	0.01179	0.01071	0.00889	0.00819	0.00260	132
S24	log1p target	boosted / bagged trees	sqrt target	0.00996	0.01235	0.01077	0.00871	0.00802	0.00275	26

code	experiment	mechanism	objective		RMSE	t+1	t+6	t+24	t+48	MAE	s
S44	scoring-aligned GRU	other / statistical	metric hour weights		0.01004	0.01248	0.01096	0.00877	0.00795	0.00321	111
I09	Tweedie emission head	other / statistical	Tweedie / Poisson		0.01022	0.01325	0.01132	0.00852	0.00777	0.00305	170
I05	archetype gate	boosted / bagged trees	plain squared error		0.01022	0.01310	0.01140	0.00856	0.00782	0.00327	79
I06	quantile two-stage	other / statistical	hurdle / two-part		0.01063	0.01238	0.01120	0.00972	0.00921	0.00331	53
I13	Poisson/Tweedie GBM	boosted / bagged trees	Tweedie / Poisson		0.01142	0.01373	0.01217	0.01013	0.00965	0.00280	27
#1											
I13	Poisson/Tweedie GBM	boosted / bagged trees	Tweedie / Poisson		0.01142	0.01373	0.01217	0.01013	0.00965	0.00280	21
#2											

F4 — Exposure (27 runs)

Contrast tested: Change which training examples are seen, holding the predictor fixed.

Prediction from the diagnosis: Robust perturbation may manufacture distinct error bases.

code	experiment	mechanism	objective		RMSE	t+1	t+6	t+24	t+48	MAE	s
A7b	S62 + hour-block dropout augmentation	BiEncSeq2Seq	dual raw+sqrt		0.00893	0.01036	0.00959	0.00824	0.00755	0.00255	170
A7	S62 + channel-dropout augmentation	BiEncSeq2Seq	dual raw+sqrt		0.00894	0.01039	0.00958	0.00824	0.00756	0.00258	143
A2b	S62 + TTA 8 views	BiEncSeq2Seq	dual raw+sqrt		0.00899	0.01038	0.00965	0.00832	0.00762	0.00257	152
A2	S62 + TTA 4 views	BiEncSeq2Seq	dual raw+sqrt		0.00900	0.01035	0.00966	0.00835	0.00762	0.00257	152
P3b	same model, no adaptation (z=0)	Seq2Seq (GRU)	plain squared error		0.00922	0.01100	0.00981	0.00834	0.00771	0.00267	121
#1											
P3 #1	test-time latent adaptation (1D)	Seq2Seq (GRU)	plain squared error		0.00922	0.01100	0.00981	0.00834	0.00771	0.00267	127
P3 #2	test-time latent adaptation (1D)	Seq2Seq (GRU)	plain squared error		0.00932	0.01115	0.00996	0.00844	0.00773	0.00273	128
P3b	same model, no adaptation (z=0)	Seq2Seq (GRU)	plain squared error		0.00932	0.01115	0.00996	0.00844	0.00773	0.00273	123
#2											
A8	S62 + prefix-truncation augmentation	BiEncSeq2Seq	dual raw+sqrt		0.00934	0.01078	0.01012	0.00865	0.00780	0.00265	214
S40	multi-origin augmentation	Seq2Seq (GRU)	plain squared error		0.00936	0.01115	0.01003	0.00847	0.00780	0.00323	225
P4	storm-phase canonicalization	Seq2Seq (GRU)	plain squared error		0.00942	0.01102	0.01002	0.00868	0.00795	0.00307	149
S35	mixup augmentation	Seq2Seq (GRU)	plain squared error		0.00944	0.01097	0.01025	0.00872	0.00781	0.00304	1088
#1											
S35	mixup augmentation	Seq2Seq (GRU)	plain squared error		0.00944	0.01097	0.01025	0.00872	0.00781	0.00304	1095
#2											
I11	storm augmentation	Seq2Seq (GRU)	plain squared error		0.00955	0.01101	0.01034	0.00877	0.00808	0.00295	1267
A9	multi-origin 47/71/103 (lull origin)	BiEncSeq2Seq	dual raw+sqrt		0.00960	0.01137	0.01024	0.00871	0.00810	0.00315	557
A9d	multi-origin 47/71/127 (wave-2 origin)	BiEncSeq2Seq	dual raw+sqrt		0.00973	0.01155	0.01043	0.00877	0.00817	0.00317	1375
A9b	multi-origin 71/103	BiEncSeq2Seq	dual raw+sqrt		0.00984	0.01195	0.01059	0.00884	0.00799	0.00326	683
S36	stochastic weight averaging	Seq2Seq (GRU)	plain squared error		0.00993	0.01215	0.01078	0.00882	0.00796	0.00324	118
#1											

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
S36	stochastic weight averaging	Seq2Seq (GRU)	plain squared error	0.00993	0.01215	0.01078	0.00882	0.00796	0.00324	116
#2										
A9c	multi-origin 47/59/71/103	BiEncSeq2Seq	dual raw+sqrt	0.00997	0.01162	0.01062	0.00909	0.00855	0.00338	958
S63b	multi-origin, 8 origins	Seq2Seq (GRU)	plain squared error	0.01008	0.01162	0.01072	0.00925	0.00873	0.00356	778
S63	multi-origin, 5 origins	Seq2Seq (GRU)	plain squared error	0.01016	0.01187	0.01077	0.00925	0.00875	0.00357	1108
S63c	multi-origin, 5 storm-phase origins	Seq2Seq (GRU)	plain squared error	0.01026	0.01208	0.01088	0.00936	0.00870	0.00333	386
M1c	MC damage only, deterministic restore	stock-flow state space	hurdle / two-part	0.01039	0.01263	0.01167	0.00921	0.00805	0.00300	205
M1b	stock-flow MC rollout, DAGger-1	stock-flow state space	hurdle / two-part	0.01052	0.01237	0.01151	0.00963	0.00857	0.00291	272
M1	stock-flow MC rollout, teacher-forced	stock-flow state space	hurdle / two-part	0.01067	0.01289	0.01194	0.00955	0.00831	0.00296	157
M1d	MC teacher-forced, 64 paths	stock-flow state space	hurdle / two-part	0.01084	0.01307	0.01211	0.00967	0.00850	0.00300	162

F5 — Estimation (23 runs)

Contrast tested: Change how the same predictor is estimated from finite data.

Prediction from the diagnosis: 191 sequences per fold implies variance-limited estimators.

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
A3x25	snapshot ensemble, 25 seeds	BiEncSeq2Seq	dual raw+sqrt	0.00862	0.01016	0.00923	0.00790	0.00720	0.00247	9244
A3	S62 + snapshot ensemble (cosine restarts)	BiEncSeq2Seq	dual raw+sqrt	0.00869	0.01009	0.00927	0.00806	0.00733	0.00250	303
A1	S62 8 seeds	BiEncSeq2Seq	dual raw+sqrt	0.00883	0.01028	0.00946	0.00816	0.00741	0.00255	346
X9	s51 dual-loss + snapshot ensembling	BiEncSeq2Seq	dual raw+sqrt	0.00884	0.01014	0.00941	0.00826	0.00755	0.00255	246
X11	i11w metric-weighted, 25 seeds	Seq2Seq (GRU)	plain squared error	0.00886	0.01039	0.00962	0.00809	0.00735	0.00268	46520
Y1	x9 (s51+snapshots) @ 25 seeds	Seq2Seq (GRU)	dual raw+sqrt	0.00890	0.01025	0.00951	0.00828	0.00758	0.00255	5859
X7	i11 storm-jitter, 25 seeds x 4 snapshots	Seq2Seq (GRU)	plain squared error	0.00892	0.01061	0.00967	0.00811	0.00730	0.00284	4515
A1x25	S62, 25 seeds	BiEncSeq2Seq	dual raw+sqrt	0.00894	0.01056	0.00958	0.00816	0.00744	0.00255	931
X3	i11 storm-jitter, 25 seeds	Seq2Seq (GRU)	plain squared error	0.00898	0.01053	0.00979	0.00819	0.00742	0.00278	11364
Y3	s35w @ 25 seeds	Seq2Seq (GRU)	plain squared error	0.00899	0.01066	0.00974	0.00818	0.00738	0.00285	15214
X6	s35 mixup + snapshot ensembling	Seq2Seq (GRU)	plain squared error	0.00902	0.01082	0.00983	0.00818	0.00724	0.00299	553
X8	s35 mixup, 25 seeds x 4 snapshots	Seq2Seq (GRU)	plain squared error	0.00905	0.01081	0.00981	0.00827	0.00732	0.00301	4530
X5	i11 storm-jitter + snapshot ensembling	Seq2Seq (GRU)	plain squared error	0.00911	0.01080	0.00985	0.00824	0.00756	0.00289	540
X4	s35 mixup, 25 seeds	Seq2Seq (GRU)	plain squared error	0.00912	0.01055	0.00983	0.00844	0.00764	0.00295	8176
X10	s51 dual-loss, 25 seeds	BiEncSeq2Seq	dual raw+sqrt	0.00924	0.01073	0.00989	0.00854	0.00782	0.00262	888
Y4	f2 @ 25 seeds	boosted / bagged trees	sqrt target	0.00930	0.01147	0.01011	0.00816	0.00746	0.00248	281
Y2	ts2c @ 25 seeds	patch / variate attention	dual raw+sqrt	0.00947	0.01163	0.01045	0.00833	0.00744	0.00278	9042
I20	GRU 8 seeds	Seq2Seq (GRU)	plain squared error	0.00948	0.01158	0.01026	0.00846	0.00764	0.00306	349
X1	s39 cross-county attention, 25 seeds	CrossCountyNet	plain squared error	0.00958	0.01169	0.01033	0.00857	0.00772	0.00303	3770
X2	s40 multi-origin, 25 seeds	Seq2Seq (GRU)	plain squared error	0.00959	0.01145	0.01023	0.00863	0.00803	0.00303	3963
Y5	p1b @ 25 seeds	stock-flow state space	hurdle / two-part	0.01042	0.01226	0.01135	0.00949	0.00857	0.00434	742
Y6	p1c @ 25 seeds	stock-flow state space	hurdle / two-part	0.01081	0.01288	0.01198	0.00975	0.00863	0.00371	349

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
Y7	r6corr @ 25 seeds	stock-flow state space	hurdle / two-part	0.01083	0.01288	0.01202	0.00986	0.00856	0.00448	787

F6 — Adaptation (7 runs)

Contrast tested: Fit a correction to or around a finished forecast.

Prediction from the diagnosis: The per-county residual has out-of-fold $R^2 \sim 0$, so post-hoc correction should fail.

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
S64	cross-fitted ensemble distillation	other / statistical	plain squared error	0.00892	0.01047	0.00944	0.00823	0.00755	0.00286	238
A4	S62 + county loss weight g=0.25	BiEncSeq2Seq	dual raw+sqrt	0.00902	0.01050	0.00969	0.00831	0.00758	0.00268	212
A4b	S62 + county loss weight g=0.50	BiEncSeq2Seq	dual raw+sqrt	0.00918	0.01072	0.00988	0.00844	0.00769	0.00282	209
M3c	mixture K=4, no MCL (collapse control)	stock-flow state space	plain squared error	0.00938	0.01122	0.01002	0.00847	0.00782	0.00266	150
M3b	mixture K=8	stock-flow state space	plain squared error	0.00966	0.01142	0.01030	0.00878	0.00815	0.00273	1514
M3	mixture-of-trajectories K=4 + MCL	stock-flow state space	plain squared error	0.00984	0.01157	0.01047	0.00901	0.00833	0.00277	493
R6corr	stock-flow + two nested flow-bias scalars	—	plain squared error	0.01090	0.01284	0.01203	0.00999	0.00874	0.00462	

F7 — Combination (4 runs)

Contrast tested: Change which trained predictors are kept and how they are pooled.

Prediction from the diagnosis: With no dominant error mode, complementary residuals should matter more than solo accuracy.

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
N1b	NCL K=4 lam=0.3	other / statistical	plain squared error	0.00920	0.01080	0.00985	0.00837	0.00776	0.00262	7029
N1c	NCL K=4 lam=0.6	other / statistical	plain squared error	0.01111	0.01353	0.01203	0.00976	0.00913	0.00339	2134
N2b	anti-corr lam=1.0	other / statistical	plain squared error	0.01187	0.01436	0.01298	0.01060	0.00953	0.00340	1337
N2	anti-corr vs shipped ensemble, lam=0.5	other / statistical	plain squared error	0.01209	0.01434	0.01321	0.01092	0.00988	0.00347	968

F8 — Uncertainty (1 runs)

Contrast tested: Change the predictive interval, not the point forecast.

Prediction from the diagnosis: Orthogonal to the point metric by construction.

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
I10	MC-dropout sampler	other / statistical	plain squared error	0.01032	0.01235	0.01113	0.00934	0.00844	0.00342	102

P — Protocol (4 runs)

Contrast tested: Procedures whose object is choosing or evaluating, not predicting.

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
RT1	ray-tuned bi-encoder (nested ASHA config)	other / statistical	plain squared error	0.00914	0.01084	0.00992	0.00829	0.00750	0.00267	183
RT3	optuna-TPE on the true metric (nested)	other / statistical	plain squared error	0.00916	0.01082	0.00988	0.00833	0.00762	0.00263	200
RT4	optuna-tuned f2 XGB config	boosted / bagged trees	sqrt target	0.00929	0.01129	0.01002	0.00821	0.00764	0.00247	16
RT2	ray-tuned on the true metric (nested)	other / statistical	plain squared error	0.00939	0.01120	0.01026	0.00851	0.00760	0.00266	159

X — Compound (5 runs)

Contrast tested: Deliberately changed more than one stage; not clean single-family evidence.

code	experiment	mechanism	objective	RMSE	t+1	t+6	t+24	t+48	MAE	s
S53	shared encoder + state residuals	other / statistical	plain squared error	0.00964	0.01119	0.01022	0.00891	0.00822	0.00325	131
S52	per-column horizon specialists	other / statistical	plain squared error	0.00970	0.01140	0.01067	0.00887	0.00786	0.00320	299
I18	in-dataset pretraining	other / statistical	plain squared error	0.01086	0.01267	0.01145	0.00983	0.00952	0.00372	184
I15	survival restoration	other / statistical	plain squared error	0.01131	0.01313	0.01225	0.01032	0.00956	0.00288	1
I01	amplitude x shape	other / statistical	plain squared error	0.01386	0.01811	0.01587	0.01137	0.01009	0.00408	2